

Kyle Chu

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EDUCATION

Cornell University | **GPA: 3.66**

Bachelor of Science in Computer Science, Minor in Artificial Intelligence

Ithaca, NY

Aug. 2023 – May 2027

RELEVANT COURSEWORK

- Data Structures
- Analysis of Algorithms
- Machine Learning
- Artificial Intelligence
- Deep Learning
- Computer Vision
- Robotics
- Systems Programming
- Operating Systems
- Computer Networks
- Functional Programming
- Text Mining

EXPERIENCE

Software Development Engineer Intern

June 2026 – August 2026

Amazon Web Services (AWS)

Seattle, WA

- Designed and shipped an end-to-end ML system using **AWS SQS, SNS, S3, and DynamoDB** that automatically identified store employees, reducing unidentified sessions across **450** stores
- Engineered the full backend communication infrastructure between **3** independent services in **Kotlin** to route **290,000+** monthly unidentified sessions through a computer vision based classifier, achieving a **10 minute** average end-to-end latency
- Architected a computer vision pipeline using deep learning embeddings and clustering through **Python** that compares an unidentified session's visual appearance against known employee images, achieving **88%** precision at a **0%** false positive rate
- Reduced manual operations review volume by **70%** by automatically reclassifying employee sessions that would otherwise require human triaging

Undergraduate Teaching Assistant | **CS 4670: Computer Vision**

January 2026 – May 2026

Cornell Ann S. Bowers College of Computing and Information Science

Ithaca, NY

- Facilitated technical mastery for **170+** students by leading office-hour sessions and grading projects on core computer-vision topics such as feature detection, epipolar geometry, and structure-from-motion, using **Python** and **OpenCV**
- Collaborated with a **20-person** staff to design CNN, semantic-segmentation, and vision-language assignments in **PyTorch** and **NumPy**, and standardized grading workflows, which improved grading consistency and reduced turnaround time

Software Engineering Intern

June 2025 – August 2025

Nonlinear

San Francisco, CA (remote)

- Automated AEC workflow execution across **15+** companies by architecting an LLM-powered, node-based AI workflow builder with **FastAPI, React, LangChain, and PostgreSQL**
- Reduced manual PDF parsing time by **70%** by shipping OCR and structured text extraction pipelines using **AWS Textract, PyMuPDF, and Pydantic**
- Built full-stack compliance tooling to reduce document errors by **63%** across **1,000+** documents through automated PDF comparison and annotation workflows using **React, Next.js, and Python**

PROJECTS

FantasyForecast | *Python (LightGBM, pandas, scikit-learn, SQLAlchemy)*

September 2025 – Present

- Developed a draft recommendation system that outperforms popular public fantasy football strategies by **19%** by training a **LightGBM** machine learning model on **12+** years of historical NFL data
- Created a start/sit weekly optimization tool that picked the best available player **67.5%** of the time by pairing **p10/p50/p90** percentile forecasts with a linear programming optimization solver
- Engineered features from raw multi-year time-series and external data sources while enforcing **144** automated tests that blocked future information from leaking into training data

Syllaparse | *TypeScript (Next.js, React), Python (FastAPI, PyMuPDF, SQLAlchemy)*

June 2025 – Present

- Built and deployed a full-stack platform for **100+** students using **OpenAI API** and **PyMuPDF** to automate lecture and exam schedule extraction from PDFs
- Integrated **Google Authentication** and **Google Calendar API**, reducing manual student data entry times by **92%** through streamlined authentication and API management
- Designed and implemented a **PostgreSQL** database and **Google Cloud Storage** to support efficient data management and faster query performance

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, Kotlin, Java, C, SQL, OCaml, HTML/CSS

AI/ML: PyTorch, LangChain, RAG architecture, Prompt engineering, OpenAI API, Scikit-learn, HuggingFace

Frameworks & Tools: React.js, Next.js, Node.js/Express.js, Astro, Flask, FastAPI, Tailwind CSS, Docker, AWS, GCP, Vercel, PostgreSQL, MongoDB, Redis, AI-assisted development (Cursor, Claude Code, Kiro, MCP)

Libraries: NumPy, Pydantic, React Redux, Pandas, DuckDB